

Beyond Speed Rankings: Application-Layer Broadband Performance in Nine Southeast Asian Countries

Chissanapun
Athiwarikanon
chissanapun.a@ku.th
Kasetsart University
Department of Computer
Science
Bangkok, Thailand

Kunanont Malayanont
kunanont.m@ku.th
Kasetsart University
Department of Computer
Science
Bangkok, Thailand

Pakkapon Pattanakul
pakkapon.p@ku.th
Kasetsart University
Department of Computer
Science
Bangkok, Thailand

Sukumal Kitisin
sukumal.i@ku.th
Kasetsart University
Department of Computer
Science
Bangkok, Thailand

ABSTRACT

Southeast Asian broadband is usually judged by headline speed-index rankings, which do not directly measure application performance. Thailand ranked 12th worldwide for fixed broadband in July 2026 [6]. A 2023 Thailand Consumer Council online survey found 81% of 2,924 mobile subscribers reporting usage problems [8]. Across nine countries, our corpus contains 990 million measurement records — 229.4M from Ookla Open Data and 760.6M from M-Lab NDT7, Q1 2023 to Q4 2025. Using Ookla province-quarter aggregates, we compute the test-weighted share of province-quarters whose means clear illustrative bandwidth and latency benchmarks for voice, HD video, UHD video and cloud gaming. Both datasets are crowdsourced and user-initiated rather than random samples. Voice and HD video clear their bars in essentially every province-quarter across all twelve quarters. Fixed broadband stratifies sharply by geography, with a 16× spread between the strongest and weakest capital regions. Against the combined cloud-gaming bar, the binding constraint flips by market: fixed broadband in Cambodia, Laos, Indonesia and Myanmar falls short mostly on bandwidth, while five mobile networks — including Singapore, where only 48.8% meet the latency bound by test weight — clear bandwidth almost universally yet fail mostly on latency. Work is ongoing on per-province modeling and on peak-hour behavior.

KEYWORDS

Internet measurement, broadband performance, Southeast Asia, Ookla, M-Lab NDT7, application requirements

1 MOTIVATION

Thailand ranked 12th worldwide for fixed broadband in July 2026 in Ookla’s Speedtest Global Index [6]. In a 2023 Thailand Consumer Council online survey, 81% of 2,924 mobile subscribers reported usage problems; slow and dropping connections were the most common complaints [8]. This self-selected survey does not estimate a population-wide problem rate. We ask how nine Southeast Asian countries perform against illustrative application benchmarks, beyond national download-speed averages.

2 DATA AND METHOD

We use two complementary open datasets covering Thailand, Vietnam, Laos, Myanmar, Cambodia, Indonesia, the Philippines, Malaysia and Singapore, over twelve quarters (Q1 2023–Q4 2025):

- **Ookla Open Data** [5] — 229,367,221 tests, published as quarterly tiles of roughly 600 m×600 m, fixed and mobile. Tests often use nearby on-net servers and parallel streams, emphasizing access-network capacity.
- **M-Lab NDT7** [4] — 760,607,387 single-stream measurement records to off-net servers, providing a complementary view of end-to-end throughput.

Ookla tiles are assigned to first-level administrative provinces by point-in-polygon against geoBoundaries ADM1 [7]. NDT7 province assignments use the cleaned records’ GADM 4.1 (2022) boundaries [2]. Province averages are test-weighted. We retain Ookla province-quarters with at least 100 tests and 5 tiles, and NDT7 province-quarters with at least 100 tests. For NDT7, tests with non-positive throughput or duration are dropped, and IP addresses are checked against ip-api.com to exclude hosting/proxy traffic. Because user-initiated speed tests are a biased sample [1], we report aggregate threshold coverage, without inferring population-wide application success. Coverage is uneven: Myanmar is missing 10.7% of fixed and 22% of mobile province-quarters, and some Vietnamese and Cambodian provinces lack mobile data in low-volume quarters — gaps we flag rather than interpolate.

We evaluate four application classes against the illustrative thresholds of Lübben and Misfeld [3]: voice (64 kbps, 200 ms), HD video streaming (5 Mbps), UHD video streaming (25 Mbps) and cloud gaming (44 Mbps, 25 ms). **Pass rate** is the test-weighted share of province-quarters (fixed and mobile separately, all twelve quarters, Q1 2023–Q4 2025) whose average download speed — and, where a latency bound applies, average latency — clears the threshold; this is an aggregate over 100+ tests per province-quarter. It measures the test weight of qualifying aggregate means, not the share of users, sessions or individual tests that meet a requirement. We use **Ookla** latency: most NDT7 tests in six countries terminate offshore (Thai tests mainly in Singapore or Hong Kong), mixing access and international transit delay. Ookla instead selects a nearby server. Its idle RTT to an often on-net server does not establish latency to an application server under load, or actual application QoE.

3 PRELIMINARY RESULTS

3.1 Voice and HD-video thresholds are widely met

The voice (64 kbps) and HD-video (5 Mbps) bandwidth bars pass in 100% of province-quarters, fixed and mobile, in every one of the

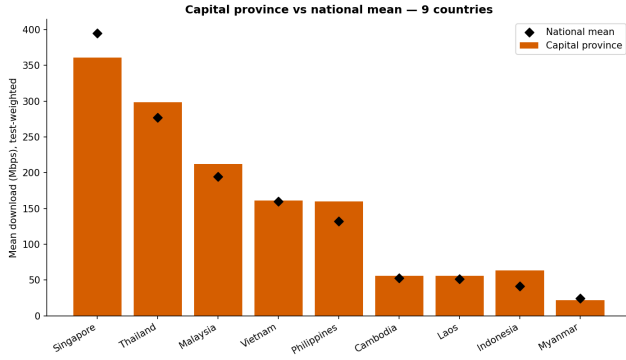


Figure 1: Capital province versus national mean download speed, fixed broadband, test-weighted, nine countries.

nine countries, across all twelve quarters. Voice’s 200 ms latency bar is nearly as uniform, with two exceptions: Myanmar’s fixed-broadband province-quarters have 98.2% test-weighted coverage (mobile: 99.1%), and Laos’s mobile province-quarters have 99.6%. These aggregate means meet modest bandwidth benchmarks; they do not establish call quality, streaming reliability or the performance of individual tests.

3.2 Fixed broadband stratifies by geography

Most capital regions outperform their own country (Figure 1): Bangkok reaches 298.8 Mbps against a national 276.9, Kuala Lumpur 212.3 against 194.8, and Manila NCR 159.6 against 132.0. Singapore and Myanmar are the exceptions: Singapore’s Central Region (361.0) sits *below* its national mean (394.8), as does Yangon (22.0 against 24.6). The spread between the strongest and weakest capital region is over 16× (Singapore’s 361.0 against Yangon’s 22.0 Mbps), a gap that a table of national averages conceals.

Absolute speed gaps remain large. Singapore nearly doubled its median provincial fixed speed over the period, from roughly 280 to 550 Mbps, and Thailand grew from roughly 200 to 300. Vietnam accelerated to join Malaysia above 200 Mbps. Cambodia and Laos both roughly doubled their median provincial speed over the period, reaching 61.7 and 62.2 Mbps by Q4 2025, and Indonesia and Myanmar reached 53.8 and 53.6 Mbps respectively — all four remain well behind the leading tier, but none is flat.

The fixed–mobile gap also varies widely. Fixed broadband exceeds mobile by 3.2× in Thailand and 2.1× in Singapore, but the two are near parity in Malaysia and Indonesia (1.1×), and Myanmar is the only country in the set where mobile is *faster* than fixed (0.8×) in these measurements.

3.3 UHD video separates the region

Raising the bar to 25 Mbps separates the region into two groups. Thailand, Vietnam, Singapore and Malaysia pass 100% of province-quarters, fixed and mobile. The Philippines is near-universal (99.6–100%), and Laos and Cambodia pass 88–99%. Indonesia and Myanmar diverge by connection type: Indonesia passes 88.1% fixed / 90.5% mobile, while Myanmar passes only 22.1% fixed but 68.8%

Table 1: Test-weighted % of province-quarters clearing each cloud-gaming component (bandwidth ≥ 44 Mbps; Ookla latency ≤ 25 ms; both), 12 quarters, Q1 2023–Q4 2025.

Country	Fixed			Mobile		
	BW	Lat	Both	BW	Lat	Both
Singapore	100.0	100.0	100.0	100.0	48.8	48.8
Vietnam	100.0	100.0	100.0	95.5	27.9	27.9
Malaysia	100.0	97.6	97.6	100.0	33.5	33.5
Thailand	100.0	99.5	99.5	99.4	21.6	21.6
Philippines	99.8	92.6	92.6	93.2	43.2	43.2
Cambodia	67.3	97.6	64.9	18.1	20.5	10.5
Laos	66.2	97.2	64.2	16.3	2.0	1.4
Indonesia	39.2	92.6	35.6	20.3	23.0	13.9
Myanmar	5.8	71.1	0.0	12.2	0.0	0.0

mobile — the one country in the set where mobile UHD threshold coverage is substantially higher than fixed, consistent with its measured fixed–mobile gap.

3.4 Cloud-gaming constraints differ by market

Table 1 shows a binding constraint that flips by market tier and connection type. In four bandwidth-constrained fixed markets — Cambodia, Laos, Indonesia and Myanmar — latency clears 71–98% by test weight, but bandwidth clears only 6–67%, capping the combined pass rate at 0–65%. On mobile, five countries — Singapore, Vietnam, Malaysia, Thailand, the Philippines — show the pattern reversed: bandwidth clears 93–100% of province-quarters, yet latency is what fails, down to just 21.6% in Thailand. Singapore is the sharpest case: mobile bandwidth coverage is 100%, but combined coverage is 48.8%. Malaysia similarly falls from 100% bandwidth coverage to 33.5% combined coverage. Download-speed comparisons miss these aggregate latency shortfalls. The other four mobile markets have low coverage on both components: Cambodia and Indonesia are slightly more bandwidth-constrained, while Laos and Myanmar have latency coverage of only 2.0% and 0.0%, alongside low bandwidth coverage.

4 ONGOING AND FUTURE WORK

Three directions are in progress. First, we are fitting province-level models of speed against economic and density covariates to examine their associations. Second, we are extending the NDT7 analysis to peak-hour behavior, where busy-versus-quiet *ratios* remain interpretable even though absolute latency is not, given offshore server placement. Third, we are quantifying the measurement bias introduced by uneven coverage and assessing sensitivity to thin-sample provinces.

Basic voice and HD-video requirements are met in almost all province-quarters. At the cloud-gaming tier, fixed networks in Cambodia, Laos, Indonesia and Myanmar are mainly bandwidth-limited, while five more advanced mobile markets are latency-limited despite sufficient bandwidth. The poster will discuss how these differences can guide targeted infrastructure improvements.

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